

MORS — Marketing Opportunity Research System

Bluhon · Technical Reference

What is MORS?

MORS is a daily intelligence system for **Bluhon**, a California public engagement, consensus building, and environmental conflict resolution firm based in the San Francisco Bay Area (est. 1995). Core services: public engagement, facilitation, mediation, consensus building, environmental conflict resolution.

MORS runs every weekday at 9:30am PT and produces a four-track intelligence report that the marketing team reads each morning to find new business opportunities.

Live at: <https://mors-runner.onrender.com>

Tech Stack

- **Backend:** Node.js (ESM — `import` syntax, not `require`) + Express
- **Hosting:** Render (mors-runner.onrender.com) — auto-deploy from GitHub `main` branch on push
- **AI:** Anthropic Claude (`claude-sonnet-4-6` via `@anthropic-ai/sdk`) with `web_search_20250305` tool (`max_uses: 40`)
- **Database:** Airtable (base ID: `appallyGF2B2bkpIU`)
- **Frontend:** Single-page HTML app at `/public/index.html` — vanilla JS, no framework
- **Scheduler:** `node-cron` — fires at `"30 9 * * 1-5"` (9:30am PT Mon-Fri)

Key Files

File	Size	Purpose
<code>mor_runner.js</code>	~2700 lines	Entire backend: scrapers, Claude calls, Airtable writes, Express routes
<code>public/index.html</code>	~1400 lines	Entire frontend: login, report viewer, pipeline, canvas, contacts
<code>public/mors.css</code>	—	Shared styles (also inlined in index.html)

Dependencies (package.json)

```
@anthropic-ai/sdk  — Claude API client
express            — HTTP server
node-cron          — Scheduler
multer             — File upload middleware
```

Airtable Tables

Constant	Table ID	Purpose
AIRTABLE_REPORTS_TABLE	tblnaSbxxkGaoscwZj	DAILY_REPORTS — one record per run
AIRTABLE_OPPTS_TABLE	tbleIosseI7FDqi9H	OPPORTUNITIES — all saved opportunities
AIRTABLE_TRACK2_TABLE	tbl4f7N5EoaKRwRXK	TRACK2_ITEMS — individual Track 2 news items
AIRTABLE_MEMORY_TABLE	tblNgcBpooPK9w0kD	PROJECT_MEMORY — learning loop patterns
AIRTABLE_SOURCES_TABLE	tblsQwva2y8ABugYH	SEARCH_SOURCES — all agency/portal URLs + firm names
AIRTABLE_MEDIA_TABLE	tblANGqT4L4Yt1MF1	MEDIA_SOURCES — Bay Area news outlets
AIRTABLE_SEARCH_QUERIES_TABLE	tblWft5ytQe3NHByq	RFP_SEARCH_QUERIES — Track 1 search query strings
AIRTABLE_KEYWORDS_TABLE	tblRKf4ftCpv1q65Z	RELEVANCE_KEYWORDS — scoring keyword weights
(hardcoded)	tblRFPCanvas	RFP_CANVAS — pursuit canvases
(hardcoded)	tblContacts	CONTACTS
(hardcoded)	tblPrimeFirms	PRIME_FIRMS
(hardcoded)	tblUsers	USERS — login authentication

DAILY_REPORTS Fields

report_date (Date), run_timestamp (Date/time), track1_html (Long text), track2_html , track3_html , track4_html

OPPORTUNITIES Fields

title , agency , deadline (Date), track (Single select), scope (Long text), source_url (URL), interest (Single select: Yes/No), geo_tier , prior_client (Checkbox), report_date (Date)

SEARCH_SOURCES Fields

source_name , url , source_type (Single select: Standalone , Portal , Prime Firm , Competitor , Governing Body), active (Checkbox), username , notes

The system reads different source_type values for different purposes:

- Standalone → agency bid page URLs passed to Claude for web_search visits (Track 1 Source B)
- Portal → procurement portal metadata shown to Claude as additional search hints
- Prime Firm / Competitor → firm names injected into Track 3/4 prompts
- Governing Body → governing body sources for Track 4

RFP_CANVAS Fields

canvas_title , active , pursuit_status , issuer_agency , rfp_url , rfp_description , proposal_due , team_notes , notes_log , files_list , created_by , created_at

Airtable API Notes

- All `atPost()` calls use `typecast: true` — required for Single Select fields, or Airtable returns 422
- Airtable Metadata API (`POST /v0/meta/bases/{baseId}/tables`) used for table creation in `/setup-airtable`
- Rate limit: batch creates use 10 records per request (Airtable max)

The Four Tracks

Track 1 — Active RFPs

Goal: find open, current RFPs/RFOs/solicitations where the scope matches Bluhon's services.

Two sources per run:

Source A — Pre-scraped portal data: 7 portal scrapers run in parallel at startup and return structured opportunities. Results are injected into the Claude prompt as a pre-scraped block with `[via:portalname]` labels.

Source B — Agency bid pages: Claude uses `web_search` to visit 60+ Bay Area agency bid pages (stored in `SEARCH_SOURCES` Airtable table, `source_type=Standalone`). Falls back to hardcoded `STANDALONE_PAGES` array in JS if Airtable returns 0 records.

Output format: HTML `<table>` saved to `track1_html` in `DAILY_REPORTS` . Columns: Agency | Solicitation # | Project/Scope | Due Date | Type | Source URL.

Persistent quality problem: Claude's `web_search` interprets page content and frequently returns expired RFPs, project pages without solicitation numbers, meeting notices, and other non-RFP content. Two mitigation layers exist but are imperfect:

1. `stripExpiredTrack1Rows(html)` — strips any `<tr>` containing a date before today
2. `parseTrack10pps(html, reportDate)` — re-filters parsed rows; drops any with a deadline before today

Portal scrapers (run in parallel via `Promise.all`):

Scraper	Site	Auth
<code>scrapeFindrfp()</code>	<code>findrfp.com</code>	Authenticated — ASP.NET form login
<code>scrapeOpengov()</code>	<code>procurement.opengov.com</code>	Authenticated
<code>scrapeBonfire()</code>	<code>gobonfire.com</code>	Authenticated — multiple Bay Area subdomains
<code>scrapePlanetbids()</code>	<code>vendorline.planetbids.com</code>	Authenticated — login OK, returns portal URLs to Claude
<code>scrapeBiddingusa()</code>	<code>biddingousa.com</code>	Authenticated — Santa Clara County, San Jose
<code>scrapeBidnet()</code>	<code>bidnetdirect.com</code>	Authenticated — Fremont, Livermore, Pleasant Hill, Novato, Tiburon, Santa Clara city, Mountain View

scrapeCivicensage()	/Bids.aspx pattern	Public — 26 Bay Area cities
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All scrapers are currently returning 0 or near-0 results. Root causes: auth flow failures, HTML structure mismatches, or portal API changes. The portal scraping pipeline is the highest-value item to fix for Track 1 reliability.

Standalone agency pages (Source B) — stored in SEARCH_SOURCES,

`source_type=Standalone` : Port of Oakland, Port of SF, BART, AC Transit, Caltrain, VTA, SMART, Golden Gate Transit, SamTrans, SFTMA, MTC/ABAG, BCDC, BAAQMD, EBRPD, MROSD, SFPUC, EBMUD, Valley Water (vendors.planetbids.com/portal/48397/bo/bo-search), Sonoma Water, Marin MWD, Zone 7 (zone7waterca.gov/construction-business-opportunities), Pleasant Hill (vendors.planetbids.com/portal/80113/bo/bo-search), all 9 Bay Area counties + SF, and ~30 cities.

Track 2 — Emerging Issues & Local Conflicts

Pre-fetched Bay Area news via RSS aggregator (`fetchAllNewsItems()`). Items are scored by relevance, deduplicated against `TRACK2_ITEMS` (7-day lookback), and the top-scored items are passed to Claude for analysis.

Claude identifies local conflicts, CEQA disputes, facility siting controversies, etc. that signal future Bluhon opportunities. Output grouped under `<h2>` headings: Local News | Regional News | Agency Board | County Board | City Board.

Each article in the frontend has a "→ **Save to Pipeline**" button.

RSS sources (all fetched in parallel):

- 24 Google News queries (topic + geography combinations)
- 21 direct feeds: Berkeleyside, Oaklandside, Mission Local, Marin Post, SJ Spotlight, Richmond Standard, East County Today, Marin IJ, Palo Alto Weekly, Mountain View Voice, Half Moon Bay Review, San Mateo Daily Journal, Press Democrat, Napa Valley Register, East Bay Times, Mercury News, SF Examiner, Novato Advance, Sacramento Bee, Planetizen, ENR

Google News URLs are resolved from redirect URLs to real article URLs via `resolveGoogleNewsUrl()` .

Track 3 — Prime Firm Activity

Contract awards, firm moves, and teaming intelligence for ~16 prime firms (AECOM, WSP, HDR, Jacobs, ICF, HNTB, Parsons, Stantec, Arup, Fehr & Peers, Kimley-Horn, GHD, EPS Group, Atkins, Burns & McDonnell, ARCADIS, Dudek).

Firm list is read from `SEARCH_SOURCES` table (`source_type=Prime Firm`) at runtime; falls back to hardcoded list.

Each article has a "→ **Save to Pipeline**" button.

Track 4 — Competitor Activity / Governing Body Pipeline

Note: The system prompt labels this "Competitor Activity" but the original intent included governing body pre-RFP signals. Current Track 4 searches for competitor firm activity (MIG, PlaceWorks, Circlepoint, Raimi+Associates, Rincon, Mintier Harnish, Dyett & Bhatia, etc.) from `SEARCH_SOURCES` (`source_type=Competitor`), with fallback to hardcoded list.

Flags ★ when RFP authorization was granted.

Claude API Architecture

Two sequential API calls per MORS run (to manage context size):

Call 1 — Tracks 1 + 2. Prompt includes:

- `SYSTEM_PROMPT` (full Bluhon MRD — ~300 lines covering services, geography, keywords, viability criteria, output format)
- `PROJECT_MEMORY` CRITICAL patterns injected dynamically
- Pre-scraped portal opps block (Source A)
- 60+ agency bid page URLs (Source B)
- Pre-fetched RSS news items (for Track 2)
- Geographic focus for the day
- Solicitation filter rules

Call 2 — Tracks 3 + 4. Prompt includes:

- Today's date + prime firm / competitor lists
- Pre-fetched Track 3 RSS items (ENR, Planetizen)
- Governing body agenda URLs
- Track 4 instructions

Claude model: `claude-sonnet-4-6` Tool: `web_search_20250305` with `max_uses: 40`

Output delimiter format (required, all six delimiters must be present):

```
---TRACK1_START--- ... ---TRACK1_END---
---TRACK2_START--- ... ---TRACK2_END---
---TRACK3_START--- ... ---TRACK3_END---
---TRACK4_START--- ... ---TRACK4_END---
---OPPORTUNITIES_JSON_START--- ... ---OPPORTUNITIES_JSON_END---
```

The JSON block contains one object per Track 1 row with fields: `title`, `agency`, `deadline` (YYYY-MM-DD or null), `track`, `scope`, `source_url`, `pursuit_type` (Prime / Sub/Team), `prior_client` (bool), `geo_tier`.

RSS Relevance Scoring

Items scored before passing to Claude. Top 60 items passed to Track 2; `when:3d` appended to all Google News queries (72-hour freshness filter).

```
KEYWORD_WEIGHTS = {
    // Tier 1 (3pts): public engagement, community engagement, facilitation,
    //   mediation, consensus, conflict resolution, community outreach, stakeholder,
    //   land use, facility siting, community opposition, neighborhood opposition,
    //   dispute, opposition, land use dispute, environmental conflict,
    //   environmental dispute, development proposal
    // Tier 2 (2pts): planning commission, environmental review, public hearing,
    //   general plan, specific plan, entitlement, water rights, board of supervisors,
    //   city council, rezoning, annexation, outreach, controversy, contested
    // Tier 3 (1pt): ceqa, eir, zoning, housing project, development project,
    //   infrastructure, advisory committee, task force, contract award, rfp
}
```

```
}  
// Title match = 2x weight. Bay Area geography match = +2pts.
```

`BAY_AREA_TERMS` : ~100 places covering all 9 counties and major cities (hardcoded in JS).

Keywords also stored in `RELEVANCE_KEYWORDS` Airtable table and read at runtime. Airtable values take precedence over hardcoded if any records exist.

Duplicate Detection

```
normalizeTitle(title) // lowercases, strips stop words (rfp, rfq, for, the, of, a,  
an, and, to, in, at, by, with, services, consulting, professional) and punctuation  
titlesMatch(a, b) // exact match OR 70%+ word overlap on words >3 chars  
fetchExistingOppTitles(cutoffStr) // pulls last 45 days from OPPORTUNITIES via  
Airtable  
isDuplicate(title) // checks cross-run (Airtable) + within-run (seenTitles Set)
```

`POSTED_CUTOFF_DAYS = 45` — used in portal scraper search queries and the Claude prompt date filter.

PROJECT_MEMORY Learning Loop

When Track 1 RFPs are reviewed in the frontend:

- "Reject" → writes a `CRITICAL:` pattern to `PROJECT_MEMORY` table
- "Pipeline" → also logs a pattern

At run time, all patterns are fetched and injected into the Claude system prompt under `QUALITY RULES LEARNED FROM USER FEEDBACK`. `CRITICAL:` patterns are injected first and most prominently.

Geographic Rotation

`getGeoFocus()` returns a `{ label, instructions }` object keyed to day of week (PT timezone):

- Mon: SF + Marin + Sonoma
 - Tue: Alameda + Contra Costa
 - Wed: Santa Clara + San Mateo
 - Thu: Napa + Solano + all special districts
 - Fri: Full Bay Area sweep + Tier 2/3
-

Prior Clients (hardcoded in prompt)

Flagged with  in all report output. List includes: ABAG, BCDC, SF Regional Water Quality Control Board, South Bay Water Recycling, U.S. EPA, Cities of Berkeley (City Manager + Parks), Half Moon Bay, Livermore, Novato, Oakland, Palo Alto, Placerville, Redwood City, San José, San Mateo, Town of Danville, Alameda County (CDA), Contra Costa County (DCD + Public Works + Supervisor Gioia), Marin County (County Executive), Santa Clara County (County Executive), Sonoma County (Supervisor Hopkins), Kaiser Permanente, Trust for Public Land, SF Bay Area Ridge Trail Council, UC Berkeley,

Frontend Architecture (public/index.html)

Single HTML file, login-gated (checks `USERS` `Airtable` table).

Pages/tabs: Today's Report | Active Pipeline | Canvas | Contacts

Key frontend functions:

Function	Purpose
<code>renderReport()</code>	Fetches latest <code>DAILY_REPORTS</code> record, renders all 4 tracks
<code>parseTrack1Html(html)</code>	Parses Claude's HTML <code><table></code> into card format with Pipeline/Reject buttons
<code>loadProspects()</code>	DISABLED — was overwriting <code>track1_html</code> content; Track 1 now renders from <code>track1_html</code> in <code>DAILY_REPORTS</code>
<code>formatTrackItems(html, track)</code>	Parses Tracks 2/3 HTML list items, adds "→ Save to Pipeline" button
<code>triggerManualRun()</code>	POSTs to <code>/run</code> (fire-and-forget), fast-polls every 15s, auto-loads when new report detected
<code>startReportPolling()</code>	Auto-polls every 90s; shows slide-up toast if newer report exists
<code>renderPipeline(opps)</code>	Renders <code>OPPORTUNITIES</code> (<code>interest=Yes</code>) as a table
<code>deleteOpportunity(id, btn)</code>	DELETES from <code>Airtable</code> + removes DOM row instantly
<code>saveArticleToPipeline(title, url, summary, track, btn)</code>	Saves Track 2/3 article to <code>OPPORTUNITIES</code> , creates Canvas
<code>createCanvas(oppId)</code>	Creates <code>RFP_CANVAS</code> record pre-populated from opportunity
<code>openCanvas(canvasId)</code>	Opens canvas detail view

Report navigation: Forward/backward date buttons at top of report view to browse historical reports by `report_date` .

Auto-poll toast: slides up from bottom — "New report ready — [Load it] ×"

Timestamps: each track header shows "Refreshed Jun 10, 2026, 9:35 AM PDT" from `run_timestamp` .

Express Routes

Route	Method	Description
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/run	POST	Fire-and-forget: starts <code>runMORSReport()</code> , responds <code>{success:true,status:'started'}</code> immediately
/test-run	POST	Abbreviated run — SF/Marin/Oakland only, Tracks 1+2 only, useful for debugging
/latest-report	GET	Returns most recent DAILY_REPORTS record
/setup-airtable	POST	One-time: creates RFP_SEARCH_QUERIES + RELEVANCE_KEYWORDS tables via Metadata API, seeds all hardcoded data
/seed-standalone-sources	POST	Seeds STANDALONE_PAGES JS array into SEARCH_SOURCES Airtable (skips existing names)
/seed-media	GET	Seeds hardcoded HARDCODED_MEDIA list into MEDIA_SOURCES Airtable
/feedback	POST	Writes user feedback (Pipeline/Reject) to PROJECT_MEMORY
/upload	POST	Multer file upload → writes to <code>/tmp/mors-files/</code> , served at <code>/files/</code>
/config	GET	Debug: returns AIRTABLE_TOKEN and base ID (should be disabled in production)
/health	GET	Returns <code>{status:'ok'}</code>
Static	—	Serves <code>/public/</code>

Cron: "30 9 * * 1-5" → 9:30am PT Mon–Fri

Environment Variables (set in Render dashboard)

ANTHROPIC_API_KEY	☒ set
AIRTABLE_TOKEN	☒ set
FINDRFP_LOGIN	☒ set
FINDRFP_PASSWORD	☒ set
OPENGOV_LOGIN	☒ set
OPENGOV_PASSWORD	☒ set
BONFIRE_LOGIN	☒ set
BONFIRE_PASSWORD	☒ set
PLANETBIDS_LOGIN	☒ set
PLANETBIDS_PASSWORD	☒ set
BIDDINGUSA_LOGIN	☒ set (also reads BIDDINGOUSA_LOGIN)
BIDDINGUSA_PASSWORD	☒ set (also reads BIDDINGOUSA_PASSWORD)
BIDNET_LOGIN	☐ account pending
BIDNET_PASSWORD	☐ account pending
CIVICENGAGE_LOGIN	☐ not needed — public pages work without auth
CIVICENGAGE_PASSWORD	☐ not needed

Airtable Helper Functions


```
atPost(tableId, fields)           // POST – creates record. typecast:true required for
Single Select fields.
atGet(tableId, params)           // GET – params is query string (e.g. ?
filterByFormula=...)
atPatch(tableId, recordId, fields) // PATCH – updates existing record
```

All three throw on non-2xx responses.

Known Issues & Unresolved Problems

1. Track 1 Reliability — CRITICAL, UNRESOLVED

Problem: Track 1 consistently returns expired RFPs, project pages without solicitation numbers, meeting notices, community gardens, fire department pages, and other non-RFP content. This has persisted across multiple fix attempts.

Root cause: Claude's `web_search` tool visits pages and uses AI interpretation to identify RFPs. This is fundamentally unreliable — Claude confuses archived solicitations for current ones, ignores date instructions, and misidentifies project descriptions as solicitations.

Mitigations in place (partial, imperfect):

- `stripExpiredTrack1Rows(html)` — strips `<tr>` elements where any date found in cell text is before today
- `parseTrack10pps(html)` — secondary filter; drops rows where parsed deadline < today
- Prompt includes: **ABSOLUTE RULE: If the deadline year is 2024 or earlier... DO NOT INCLUDE IT**
- Prompt includes: **CRITICAL SOLICITATION FILTER** with strict web-search verification rules

What would actually fix it: Reliable authenticated portal scrapers returning structured data. All 7 scrapers currently return 0 or near-0. The most valuable fix is getting FindRFP, BidNet, and/or PlanetBids returning real results. These are authenticated scrapers — auth flows may have changed.

2. Portal Scrapers All Returning 0

All 7 portal scrapers return 0 results or near-0. Known issues:

- `scrapeFindrfp()` — ASP.NET form login may be failing silently (status 302 doesn't guarantee auth success)
- `scrapePlanetbids()` — login succeeds but portal API removed; now just passes URLs to Claude
- `scrapeBidnet()` — no credentials yet
- Others — unknown, need live debugging

3. `/config` Route Exposes Token

`app.get("/config", ...)` returns `AIRTABLE_TOKEN` and `base_id` to any HTTP client. Should be removed or protected.

4. Standalone Page URLs — Some Unverified

~60 agency URLs in `SEARCH_SOURCES`. Some are pattern-guessed. Live runs log `[Standalone]` warnings for 404s — those need correction directly in Airtable (no code change required).

5. `loadProspects()` Disabled

Frontend function is disabled with `return;` at top. It was overwriting `track1_html` -rendered content with stale OPPORTUNITIES records. If Track 1 display ever needs to be re-wired to the OPPORTUNITIES table, this needs to be re-enabled carefully.

What's Built and Working

- ☒ Full four-track daily report generation via two Claude API calls
 - ☒ RSS aggregator (24 Google News queries + 21 direct feeds) with relevance scoring and URL resolution
 - ☒ 7 portal scrapers wired up and running (auth works, results sparse)
 - ☒ 60 standalone agency bid page URLs in `SEARCH_SOURCES` Airtable — read at runtime
 - ☒ Duplicate detection (within-run + 45-day cross-run)
 - ☒ `PROJECT_MEMORY` learning loop (feedback → CRITICAL patterns → injected into prompt)
 - ☒ Fire-and-forget `/run` with frontend fast-polling + auto-load
 - ☒ Auto-poll every 90s with toast notification
 - ☒ All four tracks rendering in frontend with timestamps
 - ☒ Track 1 rendered from `track1_html` in `DAILY_REPORTS` (not `OPPORTUNITIES` table)
 - ☒ `parseTrack1Html()` converts Claude's table into card format with Pipeline/Reject buttons
 - ☒ `stripExpiredTrack1Rows()` code-level expired row filter
 - ☒ "→ Save to Pipeline" button on every Track 2 and Track 3 article
 - ☒ Active Pipeline page with Canvas button + delete button
 - ☒ Historical report navigation (forward/backward date buttons)
 - ☒ RFP Canvas creation pre-populated from opportunity fields
 - ☒ File upload (`/upload`) → served at `/files/`
 - ☒ `MEDIA_SOURCES` seeded with 73 Bay Area outlets
 - ☒ Search queries, relevance keywords, firm names, and agency URLs stored in Airtable
 - ☒ Geographic rotation (Mon–Fri zones)
 - ☒ Prior client flagging (~30 warm accounts, ☒ flag in output)
 - ☒ `/setup-airtable` one-time endpoint for table creation + seeding
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What's Pending

High Priority

- **Fix portal scrapers** — FindRFP, BidNet, PlanetBids returning 0; this is the only real fix for Track 1 reliability
- **BidNet account** — user getting account; add `BIDNET_LOGIN` / `BIDNET_PASSWORD` to Render when ready
- **Remove or protect `/config` route** — exposes `AIRTABLE_TOKEN`

Medium Priority

- **Per-track refresh buttons** — separate endpoints to re-run individual tracks
- **Pipeline page** — full canvas workflow, edit fields inline
- **Contacts page** — search + edit functionality

Lower Priority

- **Verify remaining standalone URLs** — check for 404s in Render logs and correct directly in `SEARCH_SOURCES` Airtable

- **Mobile polish**
 - **Canvas → proposal workflow**
-

How to Run Locally

```
cd mors-runner
npm install
# Copy env vars from Render dashboard into .env or export them:
export ANTHROPIC_API_KEY=...
export AIRTABLE_TOKEN=...
# (etc.)
node mor_runner.js
# Open http://localhost:3000
```

Recent Git History

```
31a19d0 Restore exact original card/table format for Track 1 parsed from track1_html
      (parseTrack1Html, loadProspects disabled, report_date on opps)
      Add stripExpiredTrack1Rows filter
      Fix PlanetBids to pass portal URLs to Claude instead of direct API calls
      Fix standaloneOpps ReferenceError (removed stale reference at line 2194)
      Add report_date navigation (forward/backward date buttons in frontend)
      Add /setup-airtable endpoint – creates RFP_SEARCH_QUERIES +
      RELEVANCE_KEYWORDS tables
      Seed 60 agency URLs to SEARCH_SOURCES; system reads from Airtable at runtime
      Add typecast:true to atPost() – fixes Airtable 422 on Single Select fields
      Save to Pipeline buttons – Track 2/3 articles
      Pipeline width + trash button
      Add timestamp to track headers
      Wire BiddingUSA/BidNet/CivicEngage scrapers + 55 standalone agencies
      Update PlanetBids scraper to vendorline.planetbids.com
```